

How to select routers for homes?

Choose the router based on not only the price but also the vendor service and support and interfaces like printer ports and USB ports.

Selection should also be based on the backhaul Internet connectivity to your Wi-Fi router. If you connect through DSL broadband, it would be very important to choose a router with Wi-Fi interface plus some Ethernet ports. This type of router will cost you between Rs 2700 and Rs 4000.

If your home is connected with UTP /STP cable for the Internet, you need a Wi-Fi router without ADSL port. You can save Rs 500 to Rs 600 per router.

Ideally, you can connect 255 users per wireless access point or Wi-Fi router. However, by thumb rule, you can have 20 users with 2-3Mbps bandwidth for comfortable surfing and multimedia usage.

What are the things to consider when implementing home networks?

Choose the correct Wi-Fi device, position the router properly, change the Wi-Fi channel, increase the signal strength and range of the laptops/desktops, and ensure wireless network security.

While good Wi-Fi routers and adaptors can last many years, in general, you should periodically consider replacing your old device. Newer networking products can be faster, more reliable and offer better compatibility with your home equipment. Some people fail to consider the benefits that more advanced products like wireless print servers, gaming adaptors and video cameras can bring to their home. Before settling for a barebone network setup with only a router and a few PCs, research what else can be integrated with your Wi-Fi routers. Many options are available in the market.

Home users need to make deliberate efforts to position the Wi-Fi router such that it doesn't face any obstruction when a microwave oven or cordless phone is turned on.

In India, there are different Wi-Fi devices from different Wi-Fi standards, and there is no control on the free band, or no guidelines on how effectively this free band can be utilised. Wi-Fi equipment can transmit signals on any of several different channels, similar to TV channels. Most wireless routers ship with the same default channel number, and most home users never think about changing this. However, if you experience radio interference from a neighbour's wireless device or some other piece of electronic equipment, changing the Wi-Fi channel is often the best way to avoid it.

As with wireless routers, you can also improve the signal strength of wireless devices. Consider this when dealing with a single Wi-Fi device that suffers from a very short signal range compared to the rest of your devices. This technique can improve the ability of laptop computers to connect to Wi-Fi hotspots, for example.

Just scan the wireless network and you will find many Wi-Fi networks that are freely available to access in your

building. Many home users consider their wireless network setup a success when basic file and Internet connection sharing are functional. However, if proper security features are not in place, the job remains unfinished.

While using home routers, most security features like encryption and MAC filtering can be compromised. What's being done to tackle this problem?

One of the major reasons for these security features being compromised is the complexity involved in setting them up. It is easy for a techie, but not for a home user. In order to overcome this very complexity, there is a newer feature called WPS. WPS stands for 'Wi-Fi protected setup.' It is easy to use and implement. With this feature, the user can secure the Wi-Fi network simply by pressing a button on the Wi-Fi device. For instance, connecting your laptop to the router involves pressing the WPS button on the router and the WPS button on the laptop. Once that is done, your Wi-Fi router and Wi-Fi Laptop will be securely connected. To sum up, the WPS feature helps users of less knowhow to set up a foolproof network.

Which routers should be chosen for a home, a small company having 20 users, a medium company having 100 users and a large company having 500 users?

For home users, a broadband router with four Ethernet ports and Wi-Fi is suggested.

A router with multiple WAN ports is best for SOHO and SMB segment, where the number of users is between 20 and 100. Multiple WAN connections would provide reliability and availability. The objective is to have more than one Internet connection to the router. This would provide load balancing and redundancy. As a thumb rule, 20 users can comfortably connect on a Wi-Fi router. It is again based on the purpose of the Wi-Fi router one needs to take a decision on scalability and security of their network.

When there are more than 100 users on a Wi-Fi LAN, it is always advisable to go for managed wireless solution. There are many advantages of managed Wi-Fi LAN over multiple wireless access points/routers on the Wi-Fi LAN that are not controlled.

As more and more wireless access points/Wi-Fi routers make their way into companies, the task of managing wireless networks has grown complex. Mobile access in and around corporate office spaces and work sites is now expected, so organisations find themselves frequently adding equipment, relocating access points, modifying settings and upgrading firmware just to keep up with demand.

Most companies are using Firewall policies, but hackers are still able to break through these? What are the loopholes and how to overcome these?

It is a myth that hackers would access your network. Until you have made hidden provision for them knowingly or unknowingly, it is not possible. ●